Progressive exercises

E1 · Direct application. Determine all antiderivatives on $\mathbb{R}$ of the function $x \mapsto 3x^2 - 4x + 1$.

E2 · Application and justification. Determine the antiderivative F of the function $x \mapsto \frac{2x}{x^2+4}$ on $\mathbb{R}$ which satisfies $F(0) = 0$.

E3 · Method to organise. Determine the solution on $\mathbb{R}$ of the differential equation $y' = 3y$ which satisfies $y(1) = 2$.

E4 · Choice of strategy. Determine the solution on $\mathbb{R}$ of the differential equation $y' = -y + 4$ which satisfies $y(0) = 7$.

E5 · Synthesis or proof. Consider the differential equation $(E) : y' = y - x$ on $\mathbb{R}$. Check that $g(x) = x + 1$ is a solution of it, then find the solution of (E) which satisfies $y(0) = 0$.